



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: Signals and Systems Dept.: Electrical & Electronic Engineering
Exam Duration: 2 hours Exam Paper Setter: Weijie Yuan, Fei Chen, Fan Liu

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Score															

This exam paper contains 7 parts and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Problem 1: Choose the correct one from the four choices ($2' \times 10 = 20'$)

1. Which one of the followings is not an LTI system?

A. $y[n] = \sum_{k=n-3}^n x[k] + 3$

B. $y(t) = \int_{-3}^0 x(t - \tau) e^{\tau} d\tau$

C. $y(t) = \int_{t-3}^t x(\tau) e^{t-\tau} d\tau$

D. $y(t) = \frac{dx(t)}{dt}$

2. Which one of the following signals is aperiodic?

A. $x(t) = \sin(3t) + \cos(10t + \frac{\pi}{6})$

B. $x[n] = \sin(3n) + \cos(10\pi n)$

C. $x[n] = e^{j\frac{\pi}{2}n} + e^{j\frac{\pi}{4}n}$

D. $x(t) = 2 \cos(\frac{n\pi}{4}) + 3 \cos(\frac{n\pi}{2})$

3. A DT LTI system S includes two LTI systems (system A and system B) in cascade. The unit impulse responses are $h_1[n]$ and $h_2[n]$, respectively, for system A and system B. What is the unit impulse response of system S?

A. $h_1[n] + h_2[n]$

B. $h_1[n] \times h_2[n]$

C. convolution of $h_1[n]$ and $h_2[n]$

D. depends on the input signal

4. For the following given signals, determine which one has Fourier series?

A. $x(t) = e^{2t}u(t)$

B. $x(t) = e^t + e^{-t}$

C. $FT\{x(t)\} = \sin(\omega)$

D. $FT\{x(t)\} = \delta(\omega - 2) * \delta(\omega - 4)$

5. Consider a CT period signal $f(t)$ corresponding to the spectrum $F(j\omega)$, which one of the following is true?

A. $F(j\omega)$ is periodic and continuous

B. $F(j\omega)$ is periodic and discrete

C. $F(j\omega)$ continuous and aperiodic

D. $F(j\omega)$ is discrete and aperiodic

6. A CT system $h(t)$ has frequency response of $H(j\omega) = e^{-j3\omega}$, which one of the following statements is false?

A. The magnitude response is the same at all frequency

B. This is a linear phase system

C. This is a casual system

D. This is not a stable system

7. For a DT LTI system $y[n] = (x[n] + x[n - 1])/2$, its frequency response is equivalent to ____.

A. low-pass filter

B. band-pass filter

C. high-pass filter

D. depends on the properties of $x[n]$

8. Given the CTFT $X(j\omega)$ and DTFT $X(e^{j\omega})$ associated with real $x(t)$ and $x[n]$, respectively, which one of the following statements is false?

A. The convolution of an odd FT $X_1(j\omega)$ with an even FT $X_2(j\omega)$ is odd.

- B. If $X(e^{j\omega}) = X(e^{j(\omega-1)})$, then $x[n] = 0, |n| > 0$
 C. If $X(e^{j\omega}) = X(e^{j2\omega})$, then $x[n] = 0, |n| > 0$
 D. If $X(e^{j\omega}) = X(e^{j(\omega-2\pi)})$, then $x[n] = 0, |n| > 0$
9. A CT signal $x(t) = \cos(200\pi t) + \cos(300\pi t)$ is ideally sampled at frequency of 620π . If the sampled signal is passed through an ideal low-pass filter with a cut-off frequency of 500π . The highest frequency component in the output is ____.
- A. 420π B. 320π
 C. 280π D. 200π
10. If the ideal sampling is performed on $f(t)$ with a rate of f_s , then for sampling $f(\frac{1}{3}t - 2)$, the rate is ____, when $f(t) = \text{sinc}(100t)$, the minimum sampling rate f_s is _____. (Hint: the sampling frequency $\omega_s = 2\pi f_s$)
- A. $3f_s, \frac{100}{\pi}$ B. $\frac{1}{3}f_s, \frac{100}{\pi}$
 C. $3(f_s - 2), \frac{\pi}{200}$ D. $\frac{1}{3}(f_s - 2), \frac{200}{\pi}$

Problem 2:(10')

- a) Given $x[n] = u[n+1] - u[n-4]$, where $u[n]$ is the unit step sequence and $h[n] = \delta[n] + 2^n(u[n-1] - u[n-3])$, determine $y[n] = x[n] * h[n]$ and sketch $y[n]$.

Problem 3(6' + 4' = 10')

Given a discrete-time periodic signal $x[n] = \sin(\frac{\pi n}{4}) + 2\cos(\frac{\pi n}{2})$,

- a) Determine its period value (N) and the non-zero Fourier series coefficients from a_{-5} to a_3 .
 b) What are the values of its Fourier series coefficients a_{20} and a_{2023} ?

Problem 4 (3' + 7' + 5' = 15')

A LTI system S has the frequency response

$$H(j\omega) = \frac{3j\omega + 1}{2 - \omega^2 + 3j\omega}$$

- a) Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S .
 b) Determine the impulse response $h(t)$ of S . Is the system causal or not, and stable or not?
 c) What is the output of S when the input is $x(t) = \frac{1}{3}e^{-\frac{1}{3}t}u(t) + \frac{2}{9}te^{-\frac{1}{3}t}u(t)$?

Problem 5(5' + 5' + 5' = 15')

Determine the (inverse) Fourier transform of the following signals

- a) If $f(t) \rightleftharpoons F(j\omega)$, determine the FT of $\frac{d[f(2t-3)e^{j2t}]}{dt}$.
 b) $x(t) = [te^{-2t}\sin(4t)]u(t)$
 c) $X(e^{j\omega}) = \cos^2\omega + \sin^2(3\omega)$

Problem 6(7' + 8' = 15')

- a) Suppose we want to design a discrete-time LTI system which has the property that if the input is

$$x[n] = \left(\frac{1}{2}\right)^n u[n] - \frac{1}{4} \left(\frac{1}{2}\right)^{n-1} u[n-1],$$

then the output is

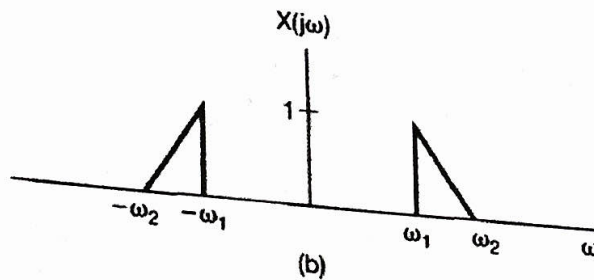
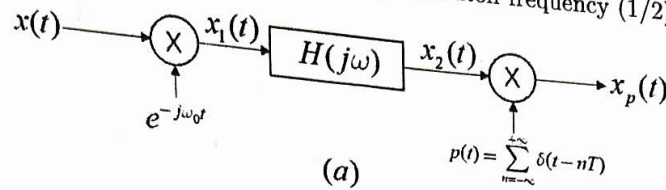
$$y[n] = \left(\frac{1}{3}\right)^n u[n]$$

Find the impulse response and frequency response, and determine a difference equation relating $x[n]$ and $y[n]$ that characterizes the system.

- b) Suppose that a system has the response $(1/4)^n u[n]$ to the input $(n+2)(1/2)^n u[n]$. If the output of this system is $\delta[n] - (-1/2)^n u[n]$, what is the input?

Problem 7(12' + 3' = 15')

Consider one sampling procedure consisting of multiplying $x(t)$ by a complex-exponential $e^{-j\omega_0 t}$ and then sampling the product. The sampling system is shown in Figure (a). With $x(t)$ real and with $X(j\omega)$ nonzero only for $\omega_1 < |\omega| < \omega_2$, the frequency of complex-exponential is chosen to be $\omega_0 = (1/2)(\omega_1 + \omega_2)$, and the ideal lowpass filter $H(j\omega)$ has cutoff frequency $(1/2)(\omega_2 - \omega_1)$.



- a) For $X(j\omega)$ as shown in Figure (b), sketch $X_1(j\omega)$, $X_2(j\omega)$, $X_p(j\omega)$.
- b) Determine the maximum sampling period T such that $x(t)$ is recoverable from $x_p(t)$.

APPENDIX

A. Formula and Definition

1. Euler Expansion: $e^{j\pm x} = \cos x \pm j \sin x$.
2. Continuous-time Convolution: $x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$
3. Discrete-time Convolution: $x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$
4. Continuous-time Fourier Series: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$, $a_k = \frac{1}{T} \int_T x(t)e^{-jk(2\pi/T)t} dt$
5. Discrete-time Fourier Series: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$, $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]e^{-jk(2\pi/N)n}$
6. Continuous-time Fourier Transform: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega$, $X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt$
7. Discrete-time Fourier Transform: $x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
8. CT Periodic Signal Fourier Transform: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$, $X(j\omega) = 2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$
9. CT Aperiodic Signal Parseval Theorem: $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega < \infty$

B. Convolution Result

1. $(u(t+T) - u(t-T)) * (u(t+T) - u(t-T)) = (2T - |t|)(u(t+2T) - u(t-2T))$ (Triangular wave!)
2. $e^{-at}u(t) * u(t) = \frac{1-e^{-at}}{a}u(t)$

C. Fourier Transform

1. $\delta(t-t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0}$ $u(t) \xrightarrow{\mathcal{F}} \frac{1}{j\omega} + \pi\delta(\omega)$ $e^{j\omega_0 t} \xrightarrow{\mathcal{F}} 2\pi\delta(\omega - \omega_0)$
2. $\cos \omega_0 t \xrightarrow{\mathcal{F}} \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ $\sin \omega_0 t \xrightarrow{\mathcal{F}} \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
3. $x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \xrightarrow{\mathcal{F}} \frac{2\sin \omega T_1}{\omega}$ $\frac{\sin Wt}{\pi t} \xrightarrow{\mathcal{F}} X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$
4. $e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{a+j\omega}$ $te^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^2}$
5. $\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^n}$
6. $\sum_{n=-\infty}^{+\infty} \delta(t-nT) \xrightarrow{\mathcal{F}} \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{T})$
7. $\delta[n-n_0] \xrightarrow{\mathcal{F}} e^{-j\omega n_0}$ $e^{j\omega_0 n} \xrightarrow{\mathcal{F}} 2\pi\delta \sum_{k=-\infty}^{+\infty} (\omega - \omega_0 - 2\pi k)$
8. $a^n u[n], |a| < 1 \xrightarrow{\mathcal{F}} \frac{1}{1-ae^{-j\omega}}$ $(n+1)a^n u[n], |a| < 1 \xrightarrow{\mathcal{F}} \frac{1}{(1-ae^{-j\omega})^2}$

D. Properties of Fourier Transform

1. CT Time Shifting and Frequency Shifting: $x(t-t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$ $e^{j\omega_0 t} x(t) \xrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$
2. CT Time and Frequency Scaling: $x(at) \xrightarrow{\mathcal{F}} \frac{1}{|a|} X(\frac{j\omega}{a})$
3. CT Differentiation and Integration: $\frac{d}{dt}x(t) \xrightarrow{\mathcal{F}} j\omega X(j\omega)$ $\int_{-\infty}^t x(\tau)d\tau \xrightarrow{\mathcal{F}} \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4. DT Time Shifting and Frequency Shifting: $x[n-n_0] \xrightarrow{\mathcal{F}} e^{-j\omega n_0} X(e^{j\omega})$ $e^{j\omega_0 n} x[n] \xrightarrow{\mathcal{F}} X(e^{j(\omega - \omega_0)})$
5. DT Difference: $x[n] - x[n-1] \xrightarrow{\mathcal{F}} (1 - e^{-j\omega})X(j\omega)$ $nx[n] \xrightarrow{\mathcal{F}} j \frac{dX(e^{j\omega})}{d\omega}$
6. Differentiation in Frequency: $tx(t) \xrightarrow{\mathcal{F}} j \frac{dX(j\omega)}{d\omega}$ $nx[n] \xrightarrow{\mathcal{F}} j \frac{dX(e^{j\omega})}{d\omega}$